SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA510

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

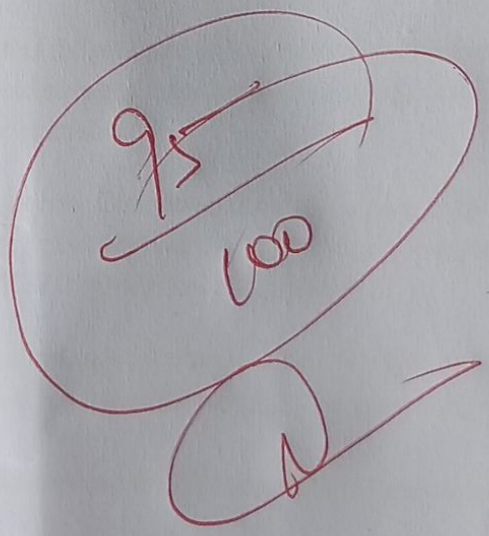
Code of Conduct:

(192524300)

I J. Lakshmi Devi *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

J. Lakshmi Devi

Signature of the student:



Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.
b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.
b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

25 25 20 15 10

95

95 100

A

CO1 Assessment tool - 01

1. Event communication security.

Scenario;

A Student organization wants to share confidential event details with its core members using a simple classical encryption method. The message is "MEET AT NOON" and a fixed shift of 4 position is used.

(a) Encryption using a shift of 4.

The method used here is the Caesar cipher. In this method, every letter is shifted a fixed number of position in the Alphabets.

Alphabets;

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

If the letters reaches Z, the alphabet starts again from A.

Given message;

MEET AT NOON

Encrypting each letter:

H	E	E	T	A	T	N	O	O	N
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
Q	I	I	X	E	X	R	S	S	R

Therefore:

MEET AT NOON

CIPHER Text: QIIX EX RSSR

Hence, the encrypted message.

The space between the word are kept uncharged.

(b) Decryption of the message.

An authorized member who knows that the encryption uses a shift of 4 can retrieve the original message by shifting every letter 4 position Backwards.

For Example:

Q I I X E X R S S R

H E E T A T N O O N

Thus,

The decryption process is the reverse.

Decryption Formula:

If letters are represented by numbers from 0 to 25

Encryption:

$$C = (P + 4) \bmod 26$$

Decryption:

$$P = (C - 4) \bmod 26$$

Where:

P = plain text.

C = Cipher text.

4 = key value.

Advantages:

* It is simple and easy to understand.
* Encryption and decryption are very fast.
* It is easy to implement in a computer program
* It is useful for learning the basic concept of cryptography.
* It does not require complex calculation.

Limitation:

1. It is not secure for modern confidential communication.
2. There are very few possible shift values.
3. An attacker can try all possible shifts.
4. Letter frequency pattern remain unchanged.
5. It can be easily broken using cryptanalysis

Conclusion:

The Caesar cipher demonstrates the basic principle of encryption. With a shift of 4, the message, "MEET AT NOON" becomes "QIIX EX RSSR". An authorized member can recover the original message by shifting each character four position backward. Although useful for educational purpose, stronger modern encryption is required for real-world security.

1. SUSPICIOUS DIGITAL ASSET INVESTIGATION

Scenario:

During a digital forensic audit, investigation suspect that a company logo may contain hidden confidential information. The image looks normal during casual inspection

(a)

The technique of hiding information inside an image is called steganography. A normal-looking image can potentially contain additional information inside its pixels, metadata or file structure.

1. Visual Inspection

First investigators can carefully, examine the image for:

* Unusual pattern.
* Strange colors.
* Unexpected noise.
* Distortions.
* Abnormal areas around edges.

2. Compare with original image.

If an original company logo is available, investigators can compare it with the suspicious

They can compare :

* Image dimensions,
* File size
* Pixel values
* Colour Information.
* Metadata :
* File Structure,

3. Examine Meta data.

Creation Information, Editing Software, Date and time, Image properties,

1. Analyze the file Structure,

Extra data, Unusual file scenarios, Unexpected headers, Embedded objects.

5. Pixel - Level Analysis

Digital images contains pixel represented by numerical values. Investigators can analyse these values for unusual pattern

6. Hash comparison.

A hash value acts like a digital fingerprint of a file.

Two techniques to strengthen the security of hidden communication.

1. Encrypt the message before embedding.

The Confidential message should first be encrypted and then embedded inside the image.

The process is:

Confidential message → Encryption → Encrypted message → Image embedding.

Example: if an attacker successfully extracts the hidden information, they will obtain encrypted the data rather than readable.

2. Use robust steganography.

A suitable steganographic technique should minimize detectable changes to the image.

* Minimize visual changes.
* Reduces obvious statistical pattern.
* Distribute hidden information appropriately.
* Preserve image quality.
* Avoid unnecessary modification.

Advantages:

1. The existence of the hidden message may not be obvious.

2. Ordinary looking images can be used as carriers.

3. It can be combined with encryption.

Limitation:

1. Advanced steganalysis may detect hidden information.

2. Image compression can affect hidden data

3. Large hidden messages may increase file size.

Conclusion:

Digital forensic investigators should not rely only on visual inspection when examining a suspicious image.